

Adam (Xinlai) Liu

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PROFILE

Lecturer and Research Associate in Cornell Systems Engineering with a research and teaching program at the intersection of **blockchain-based distributed systems**, **Trustworthy AI-driven decision-making**, and **sustainability and sociotechnical systems**. Lead instructor for *Industrial Blockchain Systems and Applications* at Cornell, project director of the SMARAG agentic-AI carbon-reporting platform, and co-investigator on **EDF–Cornell Atkinson joint research project** and **Boeing gift funding** for research and course development. Active mentor to graduate and undergraduate researchers and committed to delivering analytically rigorous, project-based graduate education for systems engineers.

APPOINTMENTS & EDUCATION

Cornell University <i>Lecturer & Research Associate, Systems Engineering</i>	July 2026
Cornell University <i>Visiting Lecturer, Systems Engineering</i>	Sep 2024 – June 2026
Cornell University <i>Ezra Systems Postdoctoral Associate, Systems Engineering</i>	Sep 2023 – June 2026
The University of Hong Kong <i>Ph.D., Industrial and Manufacturing Systems Engineering</i>	Sep 2019 – Aug 2023
Guangdong University of Technology <i>M.Eng., Mechanical Engineering</i>	Sep 2016 – Jul 2019
Shenzhen University <i>B.Eng., Mechanical Design & Manufacturing and Their Automation</i>	Sep 2012 – Jul 2016

RESEARCH INTERESTS

- **Blockchain systems** — consensus, smart contracts, tokenized incentives, traceability
- **Trustworthy AI** — physics-informed and trustworthy AI, agentic AI, retrieval-augmented generation
- **Sustainability & sociotechnical systems** — carbon markets, ESG informatics, climate accounting, circularity
- **Digital transformation** — digital twins and IoT for production and supply chain

TEACHING EXPERIENCE

Course Instructor — Cornell University

SYSEN 5450 ‘Industrial Blockchain Systems and Applications’ (graduate, on-campus and distance learning). Designed and delivered the full course covering (i) cryptography and consensus theory, (ii) Bitcoin, Ethereum, and Hyperledger Fabric platforms, and (iii) industrial applications in supply chain, sustainability, and manufacturing. The 2026 theme — *Blockchain: A Trustworthy Friend in the Era of AI* — has been **supported by Boeing Gift Funding** for student-led, project-based innovation.

Project-Course Instructor — Cornell Systems Engineering

Blockchain Project for Climate-Action Research. Have advised **more than 50 master's students** since Jan 2024 across Computer Science and Systems Engineering. Most students collaborated for two consecutive semesters, jointly producing two operational blockchain platforms, four journal papers (published), and one accepted conference paper.

Project Director — SMARAG Agentic AI Carbon-Reporting Platform

Led a five-student team in developing **SMARAG** (Self-decisive Multi-Agent Retrieval-Augmented Generation), reducing carbon-report turnaround from months to minutes. Deployed and accessed at <https://smarag.cee.cornell.edu/>. Manuscript accepted at *Expert Systems with Applications*; companion paper in preparation for *IISE Transactions*.

M.Sc. (Eng) Capstone / Dissertation Assistant Instructor — IMSE, HKU

- Jingxing Wang (2021): IoT-based environmental data collection for ESG reporting on a blockchain platform.
- Yi Wang (2021): Mobile-app-enabled data collection for blockchain-based ESG platform.
- Lin Jiang (2020): Blockchain-enabled collaborative manufacturing in 3D printing.
- Jinjin Wu (2020): Versioning smart-contract-enabled APS in smart manufacturing.

Teaching Assistant

- **IMSE4135 Systems Integration**, The University of Hong Kong, Sep 2020 – Feb 2021. Coached two undergraduate teams (10 students) on blockchain-based system development for construction and 3D-printing applications.

COURSES PREPARED TO TEACH

Mapped to the Cornell Systems Engineering Program's six emphasis areas. Course descriptions and draft syllabi available on request.

- **Industrial Blockchain Systems and Applications** — currently offered (blockchain & secure distributed systems; software-enabled systems).
- **Trustworthy AI Systems** — graduate elective on trustworthy AI principles, requirements, data, models, systems implementation. It covers multi-agentic LLM systems, retrieval-augmented generation, and trustworthy AI for engineering and policy decisions.
- **Systems Engineering for Sustainability** — quantitative carbon, ESG, and circular-economy methods; systems modeling for climate action (sustainability & sociotechnical systems).

RESEARCH FUNDING & GRANTS

EDF–Cornell Atkinson Joint Research Project (2026). *Towards Battery Circularity: Blockchain-Enabled Semantic Traceability and Incentive Design for EV Battery Recycling.* Co-investigator with Prof. Charlie Sy, Prof. H. Oliver Gao (Cornell), Dr. Maria Doa (EDF), and Abhinand Krishnashankar. Sponsored by the Environmental Defense Fund – Cornell Atkinson Center for Sustainability partnership; advances battery circularity through blockchain-enabled traceability, trustworthy AI, and supply-chain resilience.

Boeing Gift Funding (2026). *Industrial Blockchain Systems and Applications.* Course Instructor. Funds student-led, project-based innovation in the graduate course at Cornell Systems Engineering. 2026 theme: *Blockchain: A Trustworthy Friend in the Era of AI.*

Two Cornell Summer 2026 Undergraduate Research Awards (Mentor). Mentor to two awardees: (i) Joshua Dirga, *Global Climate Innovation Assessment Through the Lens of Patent Data* — supported by the Fund for Undergraduate Research on Solutions to Climate Change; and (ii) Jim Wang, *Physics-Informed Trustworthy AI for Emission Auditing of Global Renewable Energy Carbon Offset Projects* — supported by the Engaged Engineering Undergraduate Research Awards.

Cornell Systems Engineering Internal Project (2024–2026). *Agentic AI System for Carbon Reporting (SMARAG)*. Project Director; led team of five graduate students from architecture to deployment.

Hong Kong ITF Innovation and Technology Support Programme, ITP/021/20LP (Jan 2020 – Jun 2023). *nD Blockchain for ESG Reporting*. Core project member responsible for ESG market investigation, technical framework design, proposal writing, and methodology development (network topology, blockchain platform, ESG analytics).

HKU IMSE Department Project (May 2020 – Apr 2021). *imseStudio: Blockchain-Based Digital-Twin Platform for 3D-Printed Statues*. Project Manager; designed system architecture and developed two of four core modules in React.js, JSON, and Solidity.

National Natural Science Foundation of China (Sep 2016 – Aug 2017). *Multi-Scale Tool Knowledge Discovery and Reuse Based on Rough Concept Lattice Theory*. Project member; unstructured knowledge representation, inference, and matching for mold redesign.

PUBLICATIONS

Full list also available on [Google Scholar](#).

Peer-Reviewed Journal Articles

- **Liu, X.**, Pabolu, P., Jiang, Y., Fraser, T., Chen, R., & Gao, H. O. (2026). SMARAG: A Trustworthy AI System for Carbon Reporting using Self-decisive Multi-Agentic Retrieval-Augmented Generation. *Expert Systems with Applications* (Accepted). <http://dx.doi.org/10.2139/ssrn.5586584>.
- **Liu, X.**, Bagchi, T., Harish, A. R., Sy, C. L., & Gao, H. O. (2026). Blockchain and double auction for credible and efficient voluntary carbon market. *Applied Energy*, 412(1), 127719.
- **Liu, X.**, Bagchi, T., Sy, C. L., & Gao, H. O. (2026). A blockchain-based carbon registry platform for credible climate action in transportation. *npj Climate Action*, 5(1), 23.
- Ruan, S., **Liu, X.**, & Gao, H. O. (2026). Blockchain-based carbon informatics for multiple stakeholders using spatial-temporal analytics. *Environmental Modelling & Software*, 106861.
- Jiang, Y., Cheng, M., Pan, S., **Liu, X.**, Geng, Z., & Gao, H. O. (2025). Digital twin-augmented spatio-temporal Bayesian nowcasting model for emissions accounting in complex urban grid systems. *Information Sciences*, 122918.
- Harish, A. R., **Liu, X.**, Wang, X., Pan, S., Dai, H. N., Li, M., & Huang, G. Q. (2025). Blockchain for Logistics 4.0: A systematic review and prospects. *Transportation Research Part E: Logistics and Transportation Review*, 201, 104269.
- Harish, A. R., **Liu, X.**, Li, M., Zhong, R. Y., & Huang, G. Q. (2025). The new supply chain information sharing renaissance through crypto valuation mechanism of digital assets. *Transportation Research Part E: Logistics and Transportation Review*, 195, 103962.

- Niu, Y., Fu, Y., **Liu, X.**, Harish, A. R., Li, M., & Huang, G. Q. (2024). Blockchain-based incentive mechanism for environmental, social, and governance disclosure: A principal-agent perspective. *Corporate Social Responsibility and Environmental Management*, 31(6), 6318–6334.
- **Liu, X.**, Liang, W., Fu, Y., & Huang, G. Q. (2024). Dual environmental, social, and governance (ESG) index for corporate sustainability assessment using blockchain technology. *Sustainability*, 16(10), 4272.
- **Liu, X.**, Yang, Y., Fu, Y., Jiang, Y., Zhong, R. Y., Li, M., & Huang, G. Q. (2023). Data-driven ESG assessment for blockchain services: A comparative study in textiles and apparel industry. *Resources, Conservation and Recycling*, 190, 106837.
- **Liu, X.**, Jiang, Y., Wang, Z., Zhong, R. Y., Cheung, H. H., & Huang, G. Q. (2023). imseStudio: Blockchain-enabled secure digital-twin platform for service manufacturing. *International Journal of Production Research*, 61(12), 3984–4003.
- Harish, A. R., **Liu, X. L.**, Li, M., Zhong, R. Y., & Huang, G. Q. (2023). Blockchain-enabled digital assets tokenization for cyber-physical traceability in e-commerce logistics financing. *Computers in Industry*, 150, 103956.
- Jiang, Y., **Liu, X.**, Wang, Z., Li, M., Zhong, R. Y., & Huang, G. Q. (2023). Blockchain-enabled digital-twin collaboration platform for fit-out operations in modular integrated construction. *Automation in Construction*, 148, 104747.
- Kang, K., **Liu, X.**, Jiang, Y., Lee, K. K. H., Wan, S. K. W., Huang, G. Q., & Zhong, R. Y. (2023). Blockchain opportunities for the construction industry in Hong Kong: A case study of RISC and site diary. *Construction Innovation*, 23(2), 443–466.
- Wu, W., Fu, Y., Wang, Z., **Liu, X.**, Niu, Y., Li, B., & Huang, G. Q. (2022). Consortium blockchain-enabled smart ESG reporting platform with token-based incentives for corporate crowdsensing. *Computers & Industrial Engineering*, 172, 108456.
- Jiang, Y., Li, M., Li, M., **Liu, X.**, Zhong, R. Y., Pan, W., & Huang, G. Q. (2022). Digital twin-enabled real-time synchronization for planning, scheduling, and execution in precast on-site assembly. *Automation in Construction*, 141, 104397.
- **Liu, X.**, Barenji, A. V., Li, Z., Montreuil, B., & Huang, G. Q. (2021). Blockchain-based smart tracking and tracing platform for drug supply chain. *Computers & Industrial Engineering*, 161, 107669.
- Jiang, Y., **Liu, X.**, Kang, K., Wang, Z., Zhong, R. Y., & Huang, G. Q. (2021). Blockchain-enabled cyber-physical smart modular integrated construction. *Computers in Industry*, 133, 103553.
- Harish, A. R., **Liu, X. L.**, Zhong, R. Y., & Huang, G. Q. (2021). Log-Flock: A blockchain-enabled platform for digital asset valuation and risk assessment in e-commerce logistics financing. *Computers & Industrial Engineering*, 151, 107001.
- **Liu, X.**, Wang, W. M., Guo, H., Barenji, A. V., Li, Z., & Huang, G. Q. (2020). Industrial blockchain-based framework for product lifecycle management in Industry 4.0. *Robotics and Computer-Integrated Manufacturing*, 63, 101897.

Manuscripts Under Review / In Preparation

- **Liu, X.**, Fu, Y., Zhang, M., Chen, W., Harish, A. R., Shan, T. B., Zhong, R. Y., & Huang, G. Q. (2026). BESG: A Blockchain 5.0-based ESG reporting platform using generative artificial intelligence. *Humanities and Social Sciences Communications* (minor revision).

- **Liu, X.**, Liu, S., Sy, C., & Gao, H. O. (2026). A Blockchain Architecture for Trustworthy AI-enabled Battery End-of-Life Routing in Sustainable Supply Chain. Resources, Conservation and Recycling (In-preparation).
- **Liu, X.**, Liu, S., Hanumanthareddy, S., Sy, C. L., & Gao, H. O. Towards Circular Economy: A Blockchain-based Lifecycle Management Platform for Electric Vehicle Battery Recycling. Resources, Conservation and Recycling (In-preparation).

Conference Papers and Research Talks

- **Liu, X.**, Liu, S., Hanumanthareddy, S., Sy, C. L., & Gao, H. O. (2026). Repurpose or Recycle? A Blockchain Architecture for Trustworthy AI-enabled Battery End-of-Life Routing in Sustainable Supply Chain. *2026 International Conference on Resource Sustainability*, Cebu, Philippines.
- **Liu, X.**, Sy, C. L., & Gao, H. O. (2026). A Blockchain-based Electric Vehicle Battery Recycling Platform Towards Circular Economy. *IISE Annual Conference and Expo 2026*, Arlington TX, USA. (Poster and oral presentation)
- **Liu, X.**, Bagchi, T., Sy, C. L., & Gao, H. O. (2025). A double-auction mechanism for token-based carbon credit trading in a blockchain-based carbon market. *IISE Annual Conference and Expo 2025*, Atlanta, USA. (Poster and oral presentation)
- **Liu, X.**, Fu, Y., & Huang, G. Q. (2023). Data-driven dual ESG index using random forest regression. *International Conference on Resource Sustainability (icRS 2023)*, Guildford, Surrey, United Kingdom. (Poster and oral presentation)
- **Liu, X.**, Jiang, Y., & Huang, G. Q. (2022). Blockchain-based 3D printing cloud platform for service manufacturing. *Asia Pacific Industrial Engineering & Management Systems Conference* (p. 84), Taichung, Taiwan. (Poster and oral presentation)
- **Liu, X.**, Wu, H., Wu, W., Fu, Y., & Huang, G. Q. (2021). Blockchain-enabled ESG reporting framework for sustainable supply chain. In *Sustainable Design and Manufacturing 2020* (pp. 403–413), Split, Croatia. (Virtual conference)
- Li, Z., **Liu, X. L.**, & Wang, W. (2017). Simulated-annealing-algorithm-based IMMK system for mold redesign. In *Transdisciplinary Engineering: A Paradigm Shift* (pp. 889–897), Singapore.

INTELLECTUAL PROPERTY

Software Copyrights

- Toward Open Manufacturing: Blockchain-Based 3D-Printing Service System v1.0 (May 2022).
- Blockchain-Based Dual ESG Index System v1.0 (May 2022).
- Blockchain-Based Drug Traceability System v1.0 (Aug 2018).
- IoT-Based Prepackaged Food Anti-Expiration Software v1.0 (Apr 2017).
- IoT-Based Cosmetics Anti-Counterfeit System v1.0 (May 2017).

INDUSTRY & CONSULTING EXPERIENCE

Product Consultant — Shenzhen Qianhai Wearclass Technology Co., Ltd. (Apr–May 2017). Provided product-development and product-launch advisory for a precision-motor design and manufacturing firm in Shenzhen.

AWARDS & CERTIFICATES

- **Honorable Mention Young Scientist Oral Presentation Award**, 2026 International Conference on Resource Sustainability in Cebu, Philippines.
- **Cornell University**: Associate-Level CIRTl Certificate Program (2024).
- **The University of Hong Kong**: Postgraduate Scholarship (2019–2023); Teaching Assistant Certificate (2022).
- **Guangdong University of Technology**: Outstanding Graduate (2019); National “Challenge Cup” Gold Medal (2018); Academic First Scholarship (2017, 2018); Outstanding Graduate Cadres (2017, 2018); Advanced Individual of Social Work.
- **Shenzhen University**: National Inspirational Scholarship (2015–2016); Academic Honors Second Prize (2013–2014, 2014–2015); Innovation & Entrepreneurship Training Project “Excellence Award” (2014); National Grant (2013); Outstanding Member, Volunteer Assistant Branch (2013).

ACADEMIC SERVICE

Editorial Board Member, Green Technologies and Sustainability

Member, Institute of Industrial and Systems Engineers (IISE)

Conference Organization & Service:

- **Session Chair**, Circular Economy: Design, Innovation, and Industrial Systems, Sustainable Development Division, IISE Annual Conference and Expo 2026, Arlington, TX, USA.
- **Conference Volunteer**, IEEE International Conference on Automation Science and Engineering (CASE 2020), Hong Kong.

Journal Reviewer: Resources, Conservation & Recycling; International Journal of Production Economics; Scientific Reports; Technological Forecasting and Social Change; Computers & Industrial Engineering; International Journal of Production Research; Robotics and Computer-Integrated Manufacturing; International Journal of Computer Integrated Manufacturing.